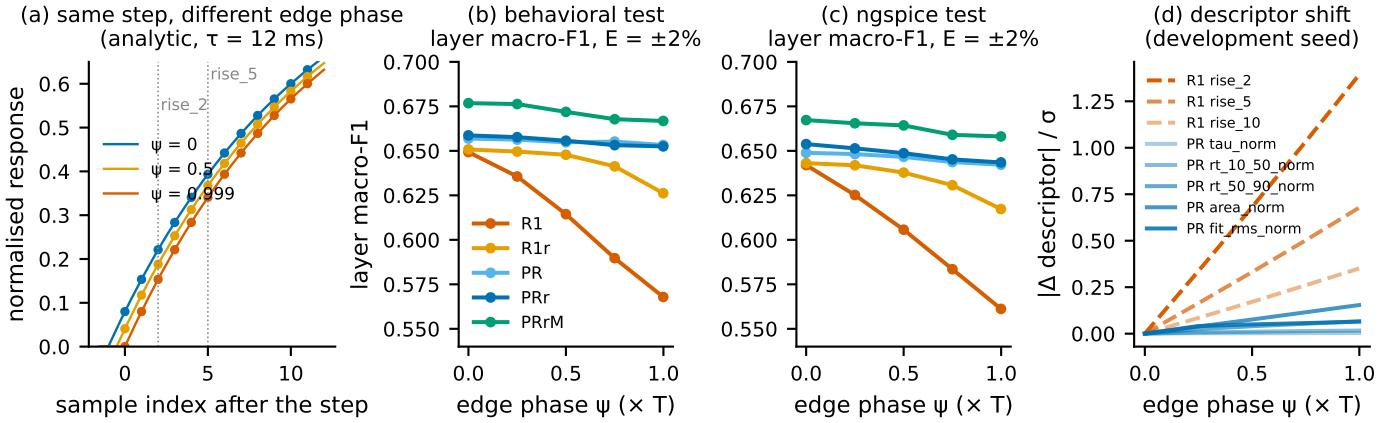
(a) same step, different edge phase
(analytic, τ = 12 ms)
normalised response
sample index after the step
rise_2
rise_5
ψ = 0
ψ = 0.5
ψ = 0.999
(b) behavioral test
layer macro-F1, E = ±2%
layer macro-F1
edge phase ψ (× T)
R1
R1r
PR
PRr
PRrM
(c) ngspice test
layer macro-F1, E = ±2%
edge phase ψ (× T)
(d) descriptor shift
(development seed)
|Δ descriptor| / σ
edge phase ψ (× T)
R1 rise_2
R1 rise_5
R1 rise_10
PR tau_norm
PR rt_10_50_norm
PR rt_50_90_norm
PR area_norm
PR fit_rms_norm